

ECS Analysis: Chemical Fixation vs Rapid HPF

Overview

Comparing extracellular space (ECS) between two tissue preparation methods (Chemical fixation vs Rapid High Pressure Freezing) across CellMap zarr groundtruth crops in four mouse tissues: Kidney, Heart, Liver, Cortex.
42 crops total, 1574 cells (normalized), 974 cells with curvature data.

Global Confound: Voxel Size is Perfectly Confounded with Prep Type

All Chemical crops are at 2nm or 4nm voxels. All HPF crops are at 8nm voxels.

There is zero overlap in resolution between the two prep types.

Prep	Voxel Sizes	N Crops
Chemical	2nm (7), 4nm (14), 8nm (3)	24
Rapid HPF	8nm (17), unknown (1)	18

Only 3 Chemical crops (Heart) share 8nm resolution with HPF. All other

Chemical crops are 2-4x higher resolution than any HPF crop.

Metrics affected: Every per-cell metric (SA:V, sphericity, curvature) shows significant correlation with voxel size (Spearman $p < 1e-4$ for all). Crop-level metrics (ECS volume fraction) are less affected since they are voxel-count ratios.

Quantified correlations (all cells, native resolution):

Metric	Spearman r vs voxel size	p-value		
SA:V ratio	-0.299	8.0e-26		
Sphericity	-0.166	1.1e-08		
Curvature median \	H\		+0.129	5.4e-05
Fraction convex	-0.071	2.8e-02		

Any Chemical vs HPF comparison is simultaneously a high-resolution vs low-resolution comparison. This cannot be deconfounded with the current data.

Metric 1: ECS Volume Fraction

Script: analyze_ecs_cell_volumes.py -> ecs_cell_volumes.csv

Plot: plot_ecs_by_prep.py -> ecs_by_prep_type.png

Method

Percentage of voxels in each crop labeled as ECS out of total crop volume.

Computed directly from segmentation label counts in zarr metadata.

Pros

- Simplest metric; just counting labeled voxels
- Least sensitive to voxel size since it is a ratio of counts
- Well-established in the EM tissue preservation literature
- Computed from metadata (fast, no heavy computation)

Cons

- One value per crop; small sample sizes (3-10 per tissue-prep group)
- No paired regions between preps
- Cannot distinguish "prep inflated ECS" from "crop landed on open space"

Known Biases and Artifacts

- Crop placement dominance: Within Chemical alone, ECS ranges from 1.4% to 74.6%. This within-group variance far exceeds any between-group signal.
- Crop size differences: Crops range from 9M to 250M voxels. Larger crops average over more tissue heterogeneity.
- Annotation completeness: crop1117 has no ECS/cell labels. Others may have incomplete annotations near crop edges.
- Tissue type confound: Each tissue has fundamentally different ECS architecture. Cross-tissue comparisons are not meaningful.

Results

Overall (all tissues pooled):

Prep	Mean ECS%	Median ECS%	IQR	N
Chemical	24.4%	20.1%	[13.1, 31.8]	24
Rapid HPF	16.0%	9.6%	[3.9, 14.7]	17

Mann-Whitney U=293, p=0.019.

By tissue:

Tissue	Chemical	Rapid HPF
Kidney	34.4 +/- 9.0% (n=7)	24.8 +/- 11.6% (n=4)
Heart	21.0 +/- 4.8% (n=3)	N/A
Liver	27.1 +/- 4.5% (n=7)	7.5 +/- 2.5% (n=10)
Cortex	13.3 +/- 3.5% (n=7)	32.9 +/- 15.0% (n=3)

Metric 2: ECS Channel Half-Width (Narrow-Filtered)

Script: analyze_ecs_width_filtered.py -> ecs_width_filtered.csv

(Original unfiltered: analyze_ecs_width.py -> ecs_width.csv)

Plot: plot_ecs_width.py -> ecs_width_by_prep.png

Method

Distance transform on ECS mask: each ECS voxel gets its distance to the nearest cell boundary (half-width). Filtered to <200nm half-width to exclude vessel lumens and large open spaces. Reports distribution stats (p10-p90, median, mean, std) plus cumulative fraction below thresholds (10-100nm).

Pros

- Measures channel geometry, not just volume
- Physical units (nm), directly interpretable
- Filtering to <200nm removes vessel/lumen contamination
- Cumulative fraction stats give richer distributional view

Cons

- Half-width, not full width; values must be doubled for literature comparison
- 200nm cutoff is somewhat arbitrary
- Local measure; does not capture connectivity or tortuosity

Known Biases and Artifacts

- Voxel quantization (MAJOR): At 8nm voxels, minimum half-width is 4nm and values are quantized to ~4nm steps. At 2nm voxels, resolution is 4x better.

This directly affects frac_below_20nm and frac_below_50nm. Demonstrated by comparing Chemical crops at different voxel sizes:

Voxel Size	N Crops	Mean Narrow Median	Mean frac<20nm
2nm (Chemical)	7	13.9 nm	0.820
4nm (Chemical)	14	29.7 nm	0.448
8nm (Chemical)	3	32.5 nm	0.338
8nm (HPF)	17	21.6 nm	0.621

The 2nm Chemical crops report narrower ECS and more sub-20nm voxels than 4nm

Chemical crops of the same prep type, demonstrating that resolution dominates the signal at fine thresholds.

- Distance transform is not true gap width: Reports distance to nearest cell surface in any direction, not specifically the gap between two opposing membranes. Overestimates width in complex geometries.
- Crop boundary contamination: ECS voxels at crop edges see artificially large distances (no cell beyond the edge).

Results (filtered <200nm)

Overall:

Prep	Mean Narrow Median	Mean frac<20nm	Mean frac<50nm	N
Chemical	25.4 nm	0.543	0.755	24
Rapid HPF	21.6 nm	0.621	0.848	17

By tissue:

Tissue	Chemical	Rapid HPF
Kidney	24.7 +/- 7.3 nm (n=7)	32.9 +/- 9.8 nm (n=4)
Heart	32.5 +/- 2.3 nm (n=3)	N/A
Liver	31.9 +/- 3.0 nm (n=7)	21.8 +/- 4.4 nm (n=10)
Cortex	13.6 +/- 8.4 nm (n=7)	30.2 +/- 16.8 nm (n=3)

Metric 3: Per-Cell SA:V Ratio

Script: analyze_cell_morphology.py -> cell_morphology_per_cell.csv

(Normalized: analyze_cell_morphology_normalized.py -> cell_morphology_normalized.csv)

Plot: plot_cell_morphology.py -> cell_morphology_by_prep.png

Method

For each segmented cell, ratio of ECS-touching surface area to cell volume.

Surface area computed by counting exposed voxel faces (face-counting on cubic grid). Only ECS-facing faces are counted (excluding cell-cell contacts and crop boundary faces). Volume is total cell voxel count x voxel volume.

Two versions exist: native resolution and normalized (all resampled to 8nm isotropic voxels). The normalized version controls for voxel-size effects on surface area.

Pros

- Per-cell metric; large sample sizes (n=1574 normalized cells)
- SA:V is dimensionless and scale-invariant in principle
- Only counts ECS-facing faces; excludes crop boundary artifacts
- Normalized version controls for voxel size

Cons

- Cells truncated at crop boundaries have distorted volumes and surfaces
- High within-group variance from cell-type diversity
- Does not directly measure ECS geometry

Known Biases and Artifacts

- Voxel staircase effect: Face-counting on a cubic grid systematically overestimates true surface area. The bias is larger at coarser voxels but is partially controlled by normalization to 8nm.

- Boundary-truncated cells (MAJOR): 601 of 1179 native-resolution cells are below 0.01 μm^3 . Many are fragments, not whole cells. 158 cells have 100% ECS-facing surface (no cell-cell contact at all), indicating isolated fragments entirely surrounded by ECS.

- SA:V direction flips with size filter: At no filter, Chemical SA:V > HPF. At $\geq 0.05 \mu\text{m}^3$ filter, Chemical SA:V < HPF. The metric is unstable with respect to the minimum cell size threshold.

- Normalization smoothing: Resampling to 8nm blurs small features, which changes SA:V and sphericity values (sphericity increases from 0.52 to 0.65 for Chemical after normalization).

Results (normalized 8nm, various size filters)

Filter	Chemical SA:V (n)	HPF SA:V (n)	p-value
None	0.050 (761)	0.043 (813)	-
$\geq 0.005 \mu\text{m}^3$	0.033 (341)	0.028 (472)	0.007
$\geq 0.01 \mu\text{m}^3$	0.023 (194)	0.025 (385)	0.033
$\geq 0.05 \mu\text{m}^3$	0.015 (90)	0.019 (230)	0.002

By tissue (native resolution, physical volume filter > 2.56M nm³):

Tissue	Chemical	Rapid HPF
Kidney	0.038 +/- 0.001 (n=230)	0.027 +/- 0.002 (n=81)
Heart	0.010 +/- 0.002 (n=11)	N/A
Liver	0.039 +/- 0.002 (n=83)	0.039 +/- 0.002 (n=142)
Cortex	0.032 +/- 0.002 (n=101)	0.033 +/- 0.001 (n=326)

Metric 4: Per-Cell Sphericity

Script: Same as Metric 3

Plot: Same as Metric 3

Method

Sphericity = $(\pi^{1/3} * (6V)^{2/3}) / A$, where V is cell volume and A is total surface area. Ranges from 0 (highly non-spherical) to 1 (perfect sphere).

Pros

- Bounded [0,1] scale; intuitive
- Per-cell metric with large sample sizes

Cons

- Closely correlated with SA:V; partially redundant
- Voxel-based surface area systematically overestimates true area, so sphericity is systematically underestimated
- "Sphericity" is a poor framing for cells that are never remotely spherical

Known Biases and Artifacts

- Voxel staircase underestimates sphericity: Manhattan geometry inflates surface area. The effect is larger at coarser voxels.
- Normalization dramatically changes values: Chemical sphericity jumps from 0.518 (native) to 0.650 (normalized 8nm) purely from resampling smoothing.
- Same boundary-truncation issues as SA:V.

Results (normalized 8nm, various size filters)

Filter	Chemical Sph. (n)	HPF Sph. (n)	p-value
None	0.650 (761)	0.601 (813)	-
$\geq 0.005 \mu\text{m}^3$	0.570 (341)	0.510 (472)	< 0.0001
$\geq 0.01 \mu\text{m}^3$	0.569 (194)	0.482 (385)	< 0.0001
$\geq 0.05 \mu\text{m}^3$	0.492 (90)	0.454 (230)	0.046

Chemical cells consistently show higher sphericity than HPF across all size filters.

Metric 5: Inter-Cell Gap Width (Voronoi)

Script: analyze_cell_contacts.py -> cell_contacts.csv

Plot: plot_cell_contacts.py -> cell_contacts_by_prep.png

Method

Voronoi tessellation of ECS: each ECS voxel assigned to nearest cell via distance transform. At Voronoi boundaries (where two cells' territories meet), gap width = sum of both cells' distance contributions. Reports median gap per crop and contact fractions at various thresholds (10-320nm).

Pros

- Directly measures cell-cell separation through ECS
- Voronoi naturally identifies cell neighbors
- Contact fraction at multiple thresholds gives cumulative distribution

Cons

- One value per crop; small sample sizes
- High within-group variance
- No per-cell pairing information

Known Biases and Artifacts

- Resolution floor (CRITICAL): At 8nm voxels, minimum measurable gap is approximately 2 x 1 voxel diagonal = 2 x 6.9nm ~ 13.8nm. HPF (all 8nm) physically cannot detect gaps below ~14nm via Voronoi. This is proven by the data:

Voxel Size	Prep	N	Mean contact_frac_20nm
2nm	Chemical	7	0.611
4nm	Chemical	14	0.223
8nm	Chemical	3	0.000
8nm	HPF	17	0.000

Chemical 8nm crops show the exact same 0% contact at 20nm as HPF 8nm crops.

The Chemical-vs-HPF difference at 20nm is entirely a voxel resolution artifact, not a biological effect.

- Open-space contamination: Vessel lumens and other large open spaces create wide Voronoi boundaries that inflate gap measurements.

Results

Overall:

Prep	Mean Gap Median	Contact frac @20nm	Contact frac @80nm	N
Chemical	48.5 nm	0.308	0.720	24
Rapid HPF	52.5 nm	0.000	0.773	17

By tissue:

Tissue	Chemical	Rapid HPF
Kidney	49.0 +/- 12.2 nm (n=7)	69.2 +/- 30.2 nm (n=4)
Heart	74.5 +/- 5.2 nm (n=3)	N/A
Liver	65.1 +/- 10.3 nm (n=7)	45.8 +/- 5.5 nm (n=10)
Cortex	20.3 +/- 6.5 nm (n=7)	52.6 +/- 23.6 nm (n=3)

Metric 6: Contact Fraction vs Distance Threshold

Script: Same as Metric 5

Plot: Same as Metric 5

Method

From the Voronoi analysis: fraction of cell-cell Voronoi boundaries with gap width below thresholds (10, 20, 40, 80, 160, 320nm).

Pros

- Richer than single median; shows gap distribution shape
- Biologically relevant thresholds (e.g., <40nm for direct signaling)

Cons

- Resolution floor makes thresholds below ~3x voxel size meaningless
- One curve per crop, small sample sizes
- Redundant with Metric 5 (same underlying data)

Known Biases and Artifacts

- Same resolution floor as Metric 5. `contact_frac_10nm` = 0% everywhere.
- `contact_frac_20nm` = 0% at 8nm voxels (all HPF and 3 Chemical crops).

Results

Tissue	Threshold	Chemical	HPF
Cortex	40nm	85.6%	53.8%
Cortex	80nm	95.2%	76.4%
Kidney	40nm	55.3%	34.0%
Kidney	80nm	72.3%	75.0%
Liver	40nm	31.0%	53.5%
Liver	80nm	56.8%	78.6%

Metric 7: Boundary-Based Contact Fraction (DISCARDED)

Script: analyze_cell_morphology.py -> cell_morphology_contacts.csv

Method

For each crop, asks: do any boundary voxels of any two cells come within X nm of each other? Reports fraction of crops where this is true.

Why Discarded

Returns 1.000 (100%) for every crop at every threshold $\geq 20\text{nm}$. Returns 0.000 for all HPF crops at 10nm and 1.000 for all Chemical crops at 10nm (pure resolution artifact). This metric has zero discriminating power. It answers "do any cells exist near each other in this crop?" which is trivially true.

Metric 8: Surface Curvature - Voxel-Based

Script: analyze_curvature.py -> cell_curvature.csv

Method

Mean curvature H at every ECS-facing surface voxel, computed via the level-set formula on a Gaussian-smoothed (sigma=16nm physical) binary cell mask. Positive H = convex (bulging into ECS), negative H = concave (invagination). Reports per-cell: median |H|, mean |H|, std H, p25/p75, fraction convex, fraction concave, median radius of curvature (1/|H|). Uses physical volume filter (> 2.56M nm³).

Pros

- Characterizes membrane shape at a local level
- Fixed physical smoothing scale (16nm) in principle makes curvature comparable across resolutions
- Per-cell metric with large sample sizes

Cons

- Curvature from finite differences on a binary voxel grid is fundamentally noisy
- Smoothing scale choice directly controls results
- Mean curvature alone does not distinguish all shape types

Known Biases and Artifacts

- Voxel staircase noise: Computing curvature from a cubic grid produces quantization artifacts. Surface normals from finite differences on binary masks are noisy.
- Smoothing scale vs resolution mismatch: 16nm = 8 voxels at 2nm but only 2 voxels at 8nm. HPF data gets less effective smoothing.
- Overwhelmingly concave (85-87%): Likely reflects that voxel-based surface normals at concave junctions are overrepresented and flat surfaces appear concave due to aliasing.
- Superseded by mesh-based curvature (Metric 9): Mesh-based computation eliminates the worst voxel staircase artifacts.

Results

By prep (all tissues, cells > 2.56M nm³):

Metric	Chemical (n=425)	HPF (n=549)		
Median H	H		0.00654 /nm	0.00523 /nm
Mean H	H		0.00731 /nm	0.00638 /nm
Median radius	153.0 nm	191.2 nm		
Fraction convex	12.7%	14.7%		
Fraction concave	87.3%	85.3%		

By tissue:

Tissue	Chemical H	HPF H	Chemical radius	HPF radius
Kidney	0.0067 (n=230)	0.0047 (n=81)	178 nm	254 nm

Heart	0.0030 (n=11)	N/A	459 nm	N/A				
Liver	0.0077 (n=83)	0.0075 (n=142)	140 nm	175 nm				
Cortex	0.0051 (n=101)	0.0053 (n=326)	221 nm	242 nm				

Metric 9: Surface Curvature - Mesh-Based

Script: analyze_curvature_mesh.py -> cell_curvature_mesh.csv

Method

Marching cubes algorithm extracts a triangulated isosurface from each cell's binary mask (with 1-voxel Gaussian smoothing). Mean curvature H computed at each mesh vertex from the Laplace-Beltrami operator on the triangle mesh. Only ECS-facing vertices are retained (vertices where the nearest non-self label is ECS rather than another cell). Reports same per-cell stats as Metric 8 plus mesh quality metadata (n_mesh_verts, n_mesh_faces, n_ecs_verts).

Pros

- Marching cubes produces a smooth triangulated surface, eliminating the worst voxel staircase artifacts
- Curvature from vertex normals on a proper mesh is geometrically more meaningful than finite differences on a binary grid
- Mesh quality metadata allows quality filtering
- Chemical vs HPF curvature is nearly identical without size filtering, suggesting this method is more stable than voxel-based

Cons

- Marching cubes + any smoothing acts as a low-pass filter; amount of smoothing determines which features survive
- Higher curvature magnitude than voxel method ($\sim 2\times$); unclear whether this reflects real features or mesh artifacts
- Computationally expensive (mesh extraction + curvature per cell)

Known Biases and Artifacts

- Implicit smoothing: Marching cubes inherently smooths the voxel surface. Additional Gaussian smoothing ($\sigma=1$ voxel) further regularizes. The net effect is resolution-dependent.
- Still concave-biased (82%): Less extreme than voxel-based (87%) but still dominated by concavity. Some concavity is real (cells wrap around each other) but the magnitude may be inflated by ECS-vertex selection bias.
- ECS-vertex selection bias: Curvature computed only at ECS-facing vertices. This preferentially samples membrane at invaginations and concavities where cells separate from neighbors, rather than flat contact surfaces.
- Mesh resolution varies with cell size: Median 26k vertices for Chemical cells vs 14k for HPF cells. HPF cells tend to be larger (higher volume in crop) but have fewer vertices due to coarser voxels. Unknown whether this affects curvature convergence.
- Method disagreement with voxel-based: Mesh median $|H|$ is $\sim 2\times$ voxel-based (0.0107 vs 0.0065 for Chemical). Mesh radius of curvature is $\sim 60\%$ smaller (94nm vs 153nm). This could mean mesh resolves real high-curvature features

or that mesh triangulation introduces noise.

Results (various size filters)

No filter:

Metric	Chemical (n=425)	HPF (n=549)		
Median \	H\		0.01066 /nm	0.01046 /nm
Mean \	H\		0.02003 /nm	0.02003 /nm
Median radius	93.8 nm	95.6 nm		
Fraction convex	18.1%	17.1%		
Fraction concave	81.9%	82.9%		
Median mesh verts	26,198	14,009		
Median ECS verts	20,190	13,481		

Size-filtered (Mann-Whitney on median |H|):

Filter	Chemical \	H\	(n)	HPF \	H\	(n)	Chemical radius	HPF radius	p-value
None	0.01066 (425)	0.01046 (549)	93.8 nm	95.6 nm	-				
>= 0.005 um3	0.00891 (340)	0.00961 (471)	112.3 nm	104.0 nm	0.086				
>= 0.01 um3	0.00659 (193)	0.00862 (383)	151.7 nm	116.0 nm	< 0.0001				
>= 0.05 um3	0.00608 (92)	0.00752 (230)	164.3 nm	132.9 nm	0.002				

Size-filtered fraction convex:

Filter	Chemical frac_convex (n)	HPF frac_convex (n)
None	0.181 (425)	0.171 (549)
>= 0.005 um3	0.244 (340)	0.206 (471)
>= 0.01 um3	0.349 (193)	0.237 (383)
>= 0.05 um3	0.418 (92)	0.319 (230)

By tissue (no size filter, values are medians across cells):

Tissue	Chemical \	H\		HPF \	H\		Chemical radius	HPF radius
Kidney	0.0129 (n=230)	0.0093 (n=81)	77.6 nm	107.7 nm				
Heart	0.0042 (n=11)	N/A	238.8 nm	N/A				
Liver	0.0130 (n=83)	0.0153 (n=142)	77.1 nm	65.5 nm				
Cortex	0.0062 (n=101)	0.0096 (n=326)	161.8 nm	104.3 nm				

Metrics Summary Table

#	Metric	Level	N (Chem/HPF)	Voxel-size confounded?	Status
1	ECS Volume Fraction	Crop	24 / 17	Low	Keep
2	ECS Half-Width (filtered)	Crop	24 / 17	High at fine thresholds	Keep with caveats
3	SA:V Ratio	Cell	761 / 813 (norm)	High ($r=-0.30$)	Keep with caveats
4	Sphericity	Cell	761 / 813 (norm)	High ($r=-0.17$)	Keep with caveats
5	Voronoi Gap Width	Crop	24 / 17	High at <40nm	Keep with caveats
6	Contact Fraction	Crop	24 / 17	Critical at <20nm	Keep at >=40nm only
7	Boundary Contact Frac.	Crop	24 / 17	-	**Discarded**
8	Curvature (voxel)	Cell	425 / 549	Moderate ($r=+0.13$)	Superseded by #9
9	Curvature (mesh)	Cell	425 / 549	Moderate	Keep

Data Sources

- Zarr crops from CellMap, accessed via SMB mount at /Volumes/cellmap/data
- Path pattern: {NRS_BASE}/{dataset}/{dataset}.zarr/recon-1/labels/groundtruth/{crop}
- Voxel sizes from zarr metadata
- ECS = semantic segmentation (0/1), Cell = instance segmentation (0=absent, nonzero=cell IDs)

Datasets

Dataset	Tissue	Prep	Voxel Size	Crops
jrc_mus-kidney	Kidney	Chemical	4nm	crop1026-1032 (7)
jrc_mus-kidney-4	Kidney	Rapid HPF	8nm	crop1134,1136,1137,1144 (4)
jrc_mus-heart-6	Heart	Chemical	8nm	crop1145-1147 (3)
jrc_mus-liver	Liver	Chemical	4nm	crop1038-1044 (7)
jrc_mus-liver-8	Liver	Rapid HPF	8nm	crop1071-1075,1123-1127 (10)
jrc_mus-cortex-2	Cortex	Rapid HPF	8nm	crop1116,1117 (2)
jrc_mus-cortex-3	Cortex	Chemical	2nm	crop1033-1037,1045,1046 (7)
jrc_mus-cortex-4	Cortex	Rapid HPF	8nm	crop1139,1141 (2)

Scripts

Script	Output	Description
`analyze_ecs_cell_volumes.py`	`ecs_cell_volumes.csv`	ECS and cell volume fractions per crop
`analyze_ecs_width.py`	`ecs_width.csv`	ECS half-width distribution (unfiltered)
`analyze_ecs_width_filtered.py`	`ecs_width_filtered.csv`	ECS half-width filtered to <200nm
`analyze_cell_morphology.py`	`cell_morphology_per_cell.csv`, `cell_morphology_crops.csv`, `cell_morphology_normalized.csv`	Per-cell SA:V, sphericity, ECS-facing SA; per-crop summaries; boundary coordinates
`analyze_cell_morphology_normalized.py`	`cell_morphology_normalized.csv`	Per-cell morphology normalized to 8nm isotropic
`analyze_cell_contacts.py`	`cell_contacts.csv`	Voronoi-based inter-cell gap widths and contact fractions
`analyze_curvature.py`	`cell_curvature.csv`	Voxel-based mean curvature per cell
`analyze_curvature_mesh.py`	`cell_curvature_mesh.csv`	Mesh-based mean curvature per cell
`plot_ecs_by_prep.py`	`ecs_by_prep_type.png`	ECS volume fraction plots
`plot_ecs_width.py`	`ecs_width_by_prep.png`	ECS width distribution plots
`plot_cell_morphology.py`	`cell_morphology_by_prep.png`	Cell morphology comparison plots
`plot_cell_contacts.py`	`cell_contacts_by_prep.png`	Cell contact distribution plots
`plot_distributions.py`	`distributions_per_crop.png`	Per-crop distribution plots
`plot_summary.py`	`ecs_analysis_summary.png`	Summary figure